

CORTEX: A FIXED-POINT THEORY OF GOVERNED CODING AGENTS*

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ABSTRACT

A *coding agent* combines a large language model (LLM) with a harness that plans, edits, and executes code. We study **Cortex**, a supervisory layer for pre-execution governance, requirement tracking, and iterative validation and repair. We formalize requirement closure on a finite lattice. For a fixed requirement set and a sound, monotone validation rule with persistent evidence, exhaustive iteration reaches the least fixed point with no more strict increases than there are requirements. Fair consequence scheduling reaches the same closure. A separate stochastic model bounds expected completion time under a uniform positive-progress assumption. These results depend on their stated assumptions; artifact repair, fallible validation, and finite execution budgets do not satisfy them automatically.

We define Gauntlet’s trajectory-similarity and build-gated scores, attack-success estimators, Wilson intervals, and case-cluster bootstrap summaries. Effective-feedback accounting counts newly certified requirements, rather than estimating their information value or utility. The five evaluation families support raw-versus-governed comparisons. The archived results have unequal observation sets, incomplete provenance, and some synthetic outcomes; they do not identify causal gains from governance or measured long-horizon scaling.

1. INTRODUCTION

An LLM coding harness turns model output into tool calls, file changes, and shell commands. Task competence and execution control are distinct properties: a capable agent can follow malicious instructions (Wei et al., 2023; Greshake et al., 2023), while a restricted agent can still leave requirements unresolved on a long task (Zhang et al., 2026).

Cortex configures governance and orchestration around a base harness. Governance applies pre-execution checks and capability/dependency policies. Orchestration analyzes instructions, tracks requirements, and coordinates validation and repair within a resource budget. Action filtering alone does not establish planning competence or complete request-level safety. Validation may use executable checks, fixed rules, or model-based judgments; their reliability is a separate evaluation question. We study these contracts through the **Gauntlet** benchmark framework.

Inference-time sampling, search, and revision trade additional computation for task performance (Snell et al., 2024; Brown et al., 2024), complementing training-time scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022). The effective-feedback perspective (Zhang et al., 2026) motivates tracking valid, nonredundant feedback that remains useful after later edits, rather than counting only tokens or tool calls. Our requirement count is an operational measure of retained progress. Classical value of information (Howard, 1966) and value of computation (Russell & Wefald, 1991) also require beliefs, utilities, and costs; a ledger update does not determine either value.

We address (i) *what an idealized supervisory layer guarantees under stated assumptions* and (ii) *how its implemented behavior can be evaluated*. The least fixed point in (i) is the closure of a specified validation rule, which may leave required work unresolved. The metrics and comparison protocol

*Canonical paper and latest results:

<https://benchmark.cortex.a2olabs.com/docs.html>

<https://benchmark.cortex.a2olabs.com/results.html>

Benchmark source: <https://github.com/Xpitfire/cortex-gauntlet>.

in (ii) do not establish that a particular implementation satisfies the theorem’s hypotheses or that historical arm differences isolate governance alone.

CONTRIBUTIONS

- A typed action-filter model and a conditional finite-lattice model of requirement closure (Sections 3–4).
- A proof of finite closure, oracle-relative soundness, and fair-schedule independence under explicit assumptions (Theorem 1), plus accounting and conditional hitting-time bounds (Propositions 3–4).
- Definitions and domain restrictions for the implemented evaluation metrics (Section 5).
- A five-family evaluation protocol, with implementation and historical-evidence limitations made explicit rather than treated as experimental conclusions (Sections 6–9).

2. RELATED WORK AND THEORETICAL LINEAGE

Neurosymbolic and cognitive computation. Combining neural generation with symbolic constraints is one form of neurosymbolic integration (Garcez & Lamb, 2023; Kautz, 2022). Validation–repair cycles also resemble the repeated rule application of classical cognitive architectures: the problem-space account of Newell and Simon (Newell & Simon, 1972), SOAR (Laird et al., 1987), and ACT-R (Anderson et al., 2004). Section 4 isolates a finite consequence operator, rather than modeling a complete cognitive architecture.

Fixed-point semantics. Our convergence argument rests on the least-fixed-point semantics of monotone operators: van Emden & Kowalski’s immediate-consequence operator and its least fixed point as the meaning of a logic program (van Emden & Kowalski, 1976), the Knaster–Tarski lattice fixed-point theorem (Tarski, 1955), and Kleene iteration (Kleene, 1952). Where an operator is a contraction on a metric space, Banach’s theorem (Banach, 1922) gives uniqueness and a rate; we use the lattice formulation because the requirement state space is naturally a finite complete lattice.

Evaluation. Capability over trajectories is scored by VERTEX, defined in Section 5.1: embedding cross-similarity against a reference descriptor set, with a presence component (BERTScore-style bidirectional matching (Zhang et al., 2020)) and an order component (distance-decayed dynamic time warping (Sakoe & Chiba, 1978)), each affinely recalibrated against the matrix’s mean-similarity baseline. Embedding comparison also appears in trajectory evaluation (Dinu et al., 2024; Patel et al., 2024). Gauntlet applies this construction to extracted descriptors, whose order need not represent an observed execution trajectory. For a positive integer k , the any-seed attack-success estimate uses the unbiased $\text{pass}@k$ estimator (Chen et al., 2021); success rates carry Wilson intervals (Wilson, 1927) and bootstrap summaries (Efron, 1979). The optional rubric judge draws on G-Eval (Liu et al., 2023), MT-Bench (Zheng et al., 2023), and Prometheus (Kim et al., 2024). Their bias analyses motivate judge validation but do not establish the validity of Gauntlet’s judges. Safety draws on the jailbreak (Wei et al., 2023) and real-world prompt-injection (Greshake et al., 2023) literatures.

Scaling, test-time compute, and effective feedback. Training scaling laws relate compute, data, and loss (Kaplan et al., 2020; Hoffmann et al., 2022); inference-time search and revision trade compute for accuracy (Snell et al., 2024; Brown et al., 2024). Effective-feedback work (Zhang et al., 2026) motivates measuring usable feedback. We count newly certified requirements, as formalized in Section 4.3. This finite-set accounting identity neither derives an empirical scaling law nor establishes that equal counts represent equal information or utility.

Iterative feedback, verification, and metareasoning. ReAct interleaves reasoning and interaction (Yao et al., 2023); Self-Refine and Reflexion study feedback and revision (Madaan et al., 2023; Shinn et al., 2023). Process supervision is a distinct approach to evaluating intermediate reasoning steps (Lightman et al., 2024). Repository issue resolution is studied in benchmarks such as SWE-bench (Jimenez et al., 2024); it is not the common evaluation setting of all these methods. Our theorem concerns any operator satisfying its assumptions. An implementation requires a separate conformance argument. Value of information (Howard, 1966) and rational metareasoning (Russell & Wefald, 1991) provide decision-theoretic context, without supplying a utility model or an optimal computation-selection policy for the present framework.

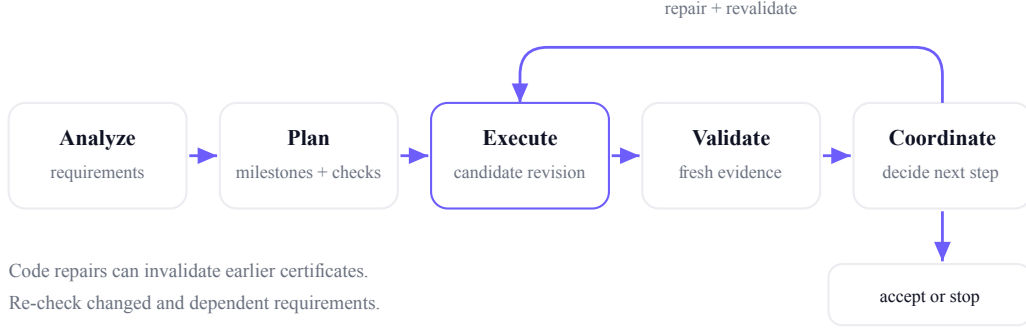


Figure 1: Schematic execution-validation-repair architecture. Its consequence-operator abstraction requires a sufficient requirement state, monotone validation, and sound persistent certificates (Section 4.2). The operational repair cycle does not establish those assumptions.

3. PRELIMINARIES AND NOTATION

We write $\mathbb{N}_0 = \{0, 1, \dots\}$, $\mathbb{N}_+ = \{1, 2, \dots\}$, $\mathbb{R}$ for the reals, and $\mathbb{I} = [0, 1]$. For finite X , $|X|$ denotes cardinality and 2^X the power set. For nonempty finite or countable X , $\Delta(X) = \{p \in \mathbb{R}_{\geq 0}^X : \sum_{x \in X} p_x = 1\}$ is the set of discrete probability distributions. For $d \in \mathbb{N}_+$, $\mathbb{S}^{d-1} = \{v \in \mathbb{R}^d : \|v\|_2 = 1\}$, with Euclidean inner product $\langle u, v \rangle \in [-1, 1]$ for $u, v \in \mathbb{S}^{d-1}$. For real x, a, b with $a \leq b$, $\text{clamp}(x; a, b) = \min(\max(x, a), b)$. The indicator $\mathbf{1}\{E\}$ equals 1 when statement E holds and 0 otherwise.

Order-theoretic objects. A complete lattice $(L, \sqsubseteq)$ has a least upper bound and a greatest lower bound for every subset, including bottom and top elements. A map $f : L \rightarrow L$ is *monotone* when $x \sqsubseteq y$ implies $f(x) \sqsubseteq f(y)$, and *inflationary* when $x \sqsubseteq f(x)$. These are distinct properties. A fixed point satisfies $f(x) = x$. A monotone map on a complete lattice has a least fixed point $\text{lfp}(f)$ (Tarski, 1955).

Agent objects. Let $\mathcal{A}$ and $\mathcal{O}$ be nonempty countable action and observation spaces. A distinguished outcome $\perp_a \notin \mathcal{A}$ denotes refusal without execution, not the lattice bottom. Set $\mathcal{A}_\perp = \mathcal{A} \cup \{\perp_a\}$ and include refusal observations in $\mathcal{O}$. The history space $\mathcal{H} = (\mathcal{A}_\perp \times \mathcal{O})^*$ contains all finite execution histories, including the empty history. A base policy $\pi : \mathcal{H} \rightarrow \Delta(\mathcal{A})$ proposes ordinary actions. For a finite, possibly empty index set J , let $h_j : \mathcal{H} \times \mathcal{A} \rightarrow \{0, 1\}$, $j \in J$, be hooks; 1 means blocked.

Task objects. Fix a finite requirement set $R = \{r_1, \dots, r_N\}$, $N \in \mathbb{N}_0$, and required subset $R_{\text{req}} \subseteq R$. The abstract state $S \subseteq R$ records certified requirements. The lattice $(2^R, \subseteq)$ has join $\cup$, meet $\cap$, bottom $\emptyset$, and top R . The abstraction $\nu : 2^R \times R \rightarrow \{0, 1\}$ determines which further certificates can be derived from S . Representing evidence by requirement identifiers alone requires the sufficiency assumption in Section 4.2; an actual workspace, tests, and execution history contain more information.

Metric objects. Let $\mathcal{D}$ be finite text descriptors and $\mathcal{D}^* = \bigcup_{m \in \mathbb{N}_0} \mathcal{D}^m$ their finite sequences. The embedding φ returns a unit vector for nonzero embeddings; the implementation uses the zero vector for a descriptor with no embedding features. Thus its full codomain is $\mathbb{S}^{d-1} \cup \{0\}$, and an exact text match need not score 1 for a zero vector. Signals $s_k \in \mathbb{I}$ combine with weights $w \in \Delta(\{1, \dots, K\})$, $K \in \mathbb{N}_+$.

4. THE CONTROL LAYER: GOVERNANCE AND ORCHESTRATION

The action filter and requirement-closure operator describe different contracts. The first restricts proposed actions at a specified history; the second describes an idealized validation closure. Neither alone proves real-world task correctness.

4.1 GOVERNANCE AS A PROJECTION ON THE ACTION SPACE

At fixed H , let $\mathcal{A}_G(H) = \{a \in \mathcal{A} : h_j(H, a) = 0 \text{ for every } j \in J\}$. Define $G_H : \mathcal{A}_\perp \rightarrow \mathcal{A}_\perp$ by

$$G_H(a) = \begin{cases} a, & a \in \mathcal{A}_G(H), \\ \perp_a, & a \notin \mathcal{A}_G(H). \end{cases}$$

In particular $G_H(\perp_a) = \perp_a$. Therefore $G_H \circ G_H = G_H$: it is an idempotent retraction onto $\mathcal{A}_G(H) \cup \{\perp_a\}$. The governed policy is the pushforward of the proposal policy:

$$\pi_G(a \mid H) = \pi(a \mid H) \mathbf{1}\{a \in \mathcal{A}_G(H)\} \quad (a \in \mathcal{A}), \quad \pi_G(\perp_a \mid H) = \sum_{a \in \mathcal{A} \setminus \mathcal{A}_G(H)} \pi(a \mid H).$$

Blocked probability becomes refusal mass, **not renormalized admissible mass**. Conditional resampling until an admissible action is obtained would be a different policy and is undefined if the original policy assigns zero mass to admissible actions. Filtering leaves accepted actions unchanged at this history, but can change later behavior and task utility.

Assumption A1 (complete mediation and hook coverage). Every effect in the claimed protection boundary is checked before execution against authoritative context, without a check/use race; every unsafe action in the specified threat class is blocked by at least one hook. Under A1 no action in that class executes. Pure predicates alone do not establish coverage, and neither false-positive rates nor semantic safety follow from idempotence. The benchmark must measure refusals and coverage separately.

4.2 ORCHESTRATION: THE COMPLETION LOOP AS A CONSEQUENCE OPERATOR

For a **fixed** oracle and requirement set define

$$\Phi(S) = S \cup \{r \in R : \nu(S, r) = 1\}.$$

Assumption A3 (oracle and evidence). Requirement identifiers are a sufficient summary for this deterministic rule. Validation is monotone: $S \subseteq T$, $\nu(S, r) = 1 \Rightarrow \nu(T, r) = 1$. Whenever the certificates in S are valid for the specified task, every new certificate $r \in R \setminus S$ with $\nu(S, r) = 1$ must be supported by valid evidence and correct for that task. Accepted transitions must preserve previously valid certificates. The union makes Φ inflationary; the additional monotonicity assumption makes it monotone. Append-only storage alone does **not** imply that assumption. For example, $R = \{a, b\}$, $\nu(S, a) = 0$ for all S , and $\nu(S, b) = \mathbf{1}\{a \notin S\}$ give $\Phi(\emptyset) = \{b\}$ but $\Phi(\{a\}) = \{a\}$.

Theorem 1 (finite closure, conditional soundness, fair-schedule independence).

Let $\Phi : 2^R \rightarrow 2^R$ be monotone and inflationary. Starting at $S_0 = \emptyset$, the iterates $S_{t+1} = \Phi(S_t)$ reach $S_* = \text{lfp}(\Phi)$ after at most $|R|$ strict increases. A further evaluation may be needed to detect stabilization. With A3's soundness and persistence conditions, all certificates in S_* remain valid. Define PASS at closure by $R_{\text{req}} \subseteq S_*$; PASS then certifies those requirements relative to the oracle, not all possible correctness properties. Starting at $\emptyset$, adding enabled consequences in any **fair** schedule reaches the same S_* .

Proof. Inflationarity gives an increasing chain. Each strict increase adds at least one of the $|R|$ elements, so the chain stabilizes. If F is any fixed point, then $S_0 \subseteq F$ and monotonicity implies $S_{t+1} = \Phi(S_t) \subseteq \Phi(F) = F$ by induction. The stabilized state is thus the least fixed point. Soundness follows by induction on certified additions, using both oracle soundness and persistence. For asynchronous addition, every added requirement is in $\Phi(S)$, so the same induction keeps the state below S_* . Fairness means no persistently enabled missing requirement is postponed forever. Since there can be only finitely many additions, the eventual state has no enabled missing requirement and is a fixed point; leastness makes it S_* . $\square$

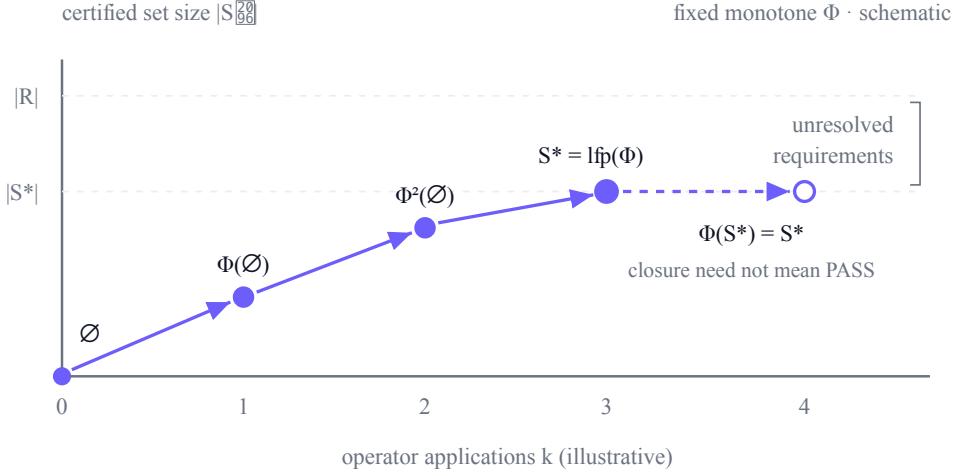


Figure 2: Illustrative Kleene ascent for the fixed monotone operator of Theorem 1, not a measured agent trace. At most $|R|$ strict increases occur; one further evaluation may detect closure. A fair schedule reaches the same least fixed point, which can leave required work unresolved.

The theorem proves derivability under ν , **not** that S_* is exactly the set of requirements achievable by arbitrary agents. Completeness of the oracle/generator would be an additional assumption. An unfair scheduler can omit enabled work forever. Detecting one unproductive *stochastic* attempt is not detecting a fixed point of a deterministic exhaustive consequence operator.

Implementation boundary. Real repair passes edit a shared artifact and can regress previously passing checks. A historical union of successful checks is not a current correctness certificate. The benchmark uses bounded retries, observation-dependent validation, and score-based candidate selection; these are not an implementation proof of A3 or exhaustive closure. Fresh validation of the final artifact is needed for PASS. Raw coding harnesses also take multiple internal actions; a single benchmark invocation is not a single consequence or a single requirement addition.

4.3 EFFECTIVE FEEDBACK AND THE VALUE OF COMPUTATION

Define the count $\text{efc}(S) = |\Phi(S) \setminus S|$ for the abstract operator. It measures newly certified requirements, motivated by retained-feedback accounts (Zhang et al., 2026). Along iteration from $\emptyset$ under A3, these additions are valid, nonredundant, and retained. Their information content and utility are not necessarily equal.

Proposition 3 (effective-feedback accounting). If $S_0 = \emptyset$ and $S_T = S_* = \text{lfp}(\Phi)$ is the stabilized iterate, then $\sum_{t=0}^{T-1} |S_{t+1} \setminus S_t| = |S_*|$. If $R_{\text{req}} \subseteq S_*$, exactly $|R_{\text{req}}|$ of these additions are required certificates. The **total** number of additions before first completion can exceed $|R_{\text{req}}|$ because optional requirements can be added too.

Proof. The disjoint differences $S_{t+1} \setminus S_t$ partition S_* . Intersecting that partition with R_{req} gives the required-only identity. $\square$

One productive pass can add any number from 1 to $|R \setminus S|$. For example, $\Phi(S) = R$ completes arbitrarily many requirements in one pass. Consequently no lower bound on required passes, no impossibility for single-invocation harnesses, and no advantage over independent sampling follows from this accounting. An append-only ledger prevents duplicate **credit**, not duplicate computation. Positive decision-theoretic value of information or net value of computation requires an explicit utility/cost model (Howard, 1966; Russell & Wefald, 1991); $\text{efc} > 0$ is not equivalent to either.

4.4 PROBABILISTIC EXECUTION OVER THE FIXED-POINT LATTICE

A stochastic process is separate from the deterministic closure model. Let $\nu_{\text{obs}} : \mathcal{H} \times 2^R \times \mathcal{A}_\perp \times \mathcal{O} \times R \rightarrow \{0, 1\}$ validate observed executions. Define $U(H, S, a, o) = S \cup \{r \in R : \nu_{\text{obs}}(H, S, a, o, r) = 1\}$ as historical certificate accumulation. A distinct current-artifact validation set may shrink. For a requirement-only, time-homogeneous Markov abstraction, **assume** the conditional law of the next requirement state depends on the past only through S . Then

$$P(S, S') = \sum_{\substack{a \in \mathcal{A}_\perp, o \in \mathcal{O} \\ U(S, a, o) = S'}} \pi_G(a \mid S) E(o \mid S, a), \quad \sum_{S' \subseteq R} P(S, S') = 1,$$

where $\pi_G(\cdot \mid S) \in \Delta(\mathcal{A}_\perp)$ and $E(\cdot \mid S, a) \in \Delta(\mathcal{O})$ is the observation distribution, including a refusal observation for $\perp_a$. Each of U, π_G, E must admit this sufficient-state representation. For general history-dependent execution use $X_t = (H_t, S_t)$ instead, including all relevant environment state in the history. Aggregating its transitions by S alone does not in general produce a Markov chain. Historical certificate accumulation gives $P(S, S') = 0$ unless $S \subseteq S'$ but does not prove current-artifact correctness.

Completion as a hitting time. Let $C = \{S \subseteq R : R_{\text{req}} \subseteq S\}$ and $T_C = \inf\{t \in \mathbb{N}_0 : S_t \in C\}$, with $\inf \emptyset = \infty$. For an increasing process C is a closed set; make its states absorbing if the run stops at first completion. Specify the initial state S_0 for all probabilities and expectations.

Proposition 4 (conditional completion time). For a finite increasing Markov chain, suppose every reachable $S \notin C$ has $\Pr(S_{t+1} \supsetneq S_t \mid S_t = S) \geq \varepsilon$, where $0 < \varepsilon \leq 1$. Then $\Pr(T_C < \infty) = 1$ and $\mathbb{E}[T_C] \leq (|R| - |S_0|)/\varepsilon$.

Proof. Since $R \in C$, at most $|R| - |S_0|$ strict increases can precede completion. Before completion, the wait for each increase is dominated by a geometric variable of mean $1/\varepsilon$. Their sum bounds T_C and its expectation; independence of the holding times is unnecessary. A finite expectation implies almost-sure completion. $\square$

An incomplete absorbing state violates the progress assumption. Without that assumption, completion need not occur and $\mathbb{E}[T_C]$ may be infinite. Governance changes the execution law through the refusal-mass filter of Section 4.1; comparing unsafe-event probabilities additionally requires an explicit unsafe-event indicator on the augmented state, not merely a subset of satisfied requirements. A difference of expected completion times is meaningful only when both expectations are finite. No transition probabilities or completion-time estimates are supplied by the lattice theorem.

5. EVALUATION

5.1 VERTEX TRAJECTORY SIMILARITY

This implementation of VERTEX combines descriptor cross-similarity with an order penalty, inspired by trajectory evaluation (Dinu et al., 2024; Patel et al., 2024). For nonempty sequences $c = (c_1, \dots, c_m)$ and $r = (r_1, \dots, r_n)$, $m, n \in \mathbb{N}_+$, define $\mathbf{S}_{ij} = \langle \varphi(c_i), \varphi(r_j) \rangle \in [-1, 1]$. For unit embeddings this is cosine similarity; zero-vector handling is as in Section 3. Either empty sequence receives score 0 by convention. Figure 3 shows the two components.

Presence (unordered). A BERTScore-style bidirectional best match (Zhang et al., 2020), where P (precision) measures how well each candidate descriptor is matched by some reference and Q (recall) how well each reference descriptor is matched by some candidate:

$$\begin{aligned}
P &= \frac{1}{m} \sum_{i=1}^m \max(0, \max_j \mathbf{S}_{ij}), \\
Q &= \frac{1}{n} \sum_{j=1}^n \max(0, \max_i \mathbf{S}_{ij}), \\
F &= \begin{cases} 2PQ/(P+Q), & P+Q > 0, \\ 0, & P+Q = 0. \end{cases}
\end{aligned}$$

Thus $P, Q, F \in \mathbb{I}$ and $\min(P, Q) \leq F \leq \max(P, Q)$. Negative similarities still enter the order cost and the matrix-mean baseline below.

Order. Dynamic time warping (Sakoe & Chiba, 1978) minimizes over paths Γ from $(1, 1)$ to (m, n) with steps $(1, 0)$, $(0, 1)$, or $(1, 1)$. For finite $\lambda \geq 0$, the cost is $d(i, j) = (1 - \mathbf{S}_{ij})(1 + \lambda|(i-1)/m - (j-1)/n|)$:

$$\text{DTW} = \min_{\gamma \in \Gamma} \sum_{(i,j) \in \gamma} d(i, j), \quad D = \max\left(0, 1 - \frac{\text{DTW}}{m+n}\right) \in \mathbb{I},$$

The path has between $\max(m, n)$ and $m + n - 1$ entries. The denominator $m + n$ is a chosen normalization, not the path length or an upper bound on the **cost**.

Matrix-mean recalibration. For $0 < \epsilon < 1$, let $b = \text{clamp}((mn)^{-1} \sum_{i,j} \mathbf{S}_{ij}; 0, 1 - \epsilon)$ and $\eta_b(x) = \text{clamp}((x - b)/(1 - b); 0, 1)$. The raw mean lies in $[-1, 1]$, and the clamp ensures a positive denominator. It is the mean of this candidate-reference pair, **not** an independently estimated null expectation. For finite $\alpha \in [0, 1]$:

$$\text{VERTEX}(c, r) = \alpha \eta_b(F) + (1 - \alpha) \eta_b(D) \in \mathbb{I}.$$

The benchmark configuration fixes $\alpha = 0.6$, $\lambda = 0.5$, and $\epsilon = 0.01$ (the baseline-clamp margin). Proposition 1 records the range and the exact sense of the calibration.

Proposition 1 (range and anchors). VERTEX lies in $\mathbb{I}$. The normalizer is nondecreasing on $\mathbb{R}$ and strictly increasing on $[b, 1]$, with $\eta_b(b) = 0$ and $\eta_b(1) = 1$. Identical sequences of nonzero unit embeddings score 1 in exact arithmetic.

Proof. All floored best matches lie in $\mathbb{I}$; the harmonic mean lies between its arguments. Costs are nonnegative, hence the clamped D lies in $\mathbb{I}$. The normalizer has the stated endpoints and slope $1/(1 - b) > 0$ on its unclamped interval. The mixture is convex. Identical unit-vector sequences have unit diagonal similarities, $P = Q = 1$, and a zero-cost diagonal path, so $F = D = 1$. $\square$

There is **no universal chance floor**. A constant zero similarity matrix with $m = n$ has $F = 0$, $D = 1/2$, hence score 0.2 at the default mixture. With $m = 1$, $n = 9$ and $\lambda = 0.5$, the same zero similarities give path cost 11, $D = 0$, and score 0. Lengths, order, and embedding geometry therefore affect unrelated-input scores. Maxima need not strictly exceed a mean, and the DTW term need not concentrate near $(1 + b)/2$. Comparisons require fixed extraction and embedding protocols; they are similarity comparisons, not calibrated probabilities or correctness certificates.

5.1.1 VERTEX-QE: ESTIMATED REFERENCES

The project task can compare a candidate with hand-authored capability and architecture descriptors. These references select particular realizations of an open-ended brief. VERTEX-QE instead estimates descriptors from public sources. Each invocation of the Section 5.1 kernel retains the same presence score F , order score D , and matrix-mean recalibration η_b ; its reference argument becomes an estimate $\hat{r}$. The project-level mixture also changes its architecture weight, as defined below.

Capability descriptors come from the public brief’s enumerated outcomes. Architecture descriptors combine engineering hints in that brief with the shipped `arch_corpus.json` vocabulary. The

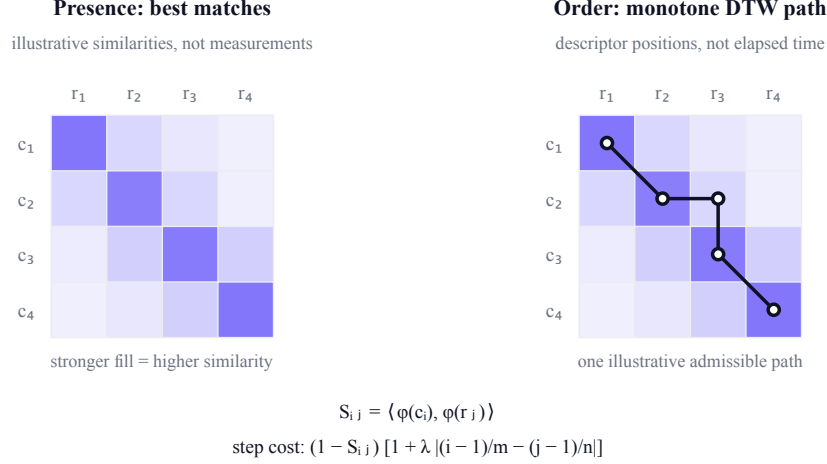


Figure 3: The two components of the VERTEX trajectory score (Section 5.1). Left: the cross-similarity matrix $S_{ij} = \langle \varphi(c_i), \varphi(r_j) \rangle$ between candidate descriptors c_i (rows) and reference descriptors r_j (columns); stronger color indicates higher similarity in this illustrative matrix. The *presence* component reads bidirectional best matches off this matrix (which reference behaviors appear, and how strongly). Right: the *order* component is a distance-decayed alignment path through the matrix whose step cost $(1 - S_{ij})(1 + \lambda |\frac{i-1}{m} - \frac{j-1}{n}|)$ penalizes matches that occur out of order. Both components are recalibrated against the mean-similarity baseline (Proposition 1).

latter contains synthetic canonical patterns, not observations from curated repositories. Empty strings and exact duplicates are removed within each source; the union greedily drops later descriptors whose cosine similarity to a retained one is at least 0.92. The resulting list has a deterministic order that affects DTW, although it does not describe execution time or a distribution of valid architectural alternatives.

For the architecture mixture, let L be the number of nonempty cleaned source lists. Write list ℓ as $(x_{\ell 1}, \dots, x_{\ell n_\ell})$, where $n_\ell \geq 1$. For distinct sources $\ell, k \in \{1, \dots, L\}$, directed agreement is

$$a_{\ell k} = \frac{1}{n_\ell} \sum_{i=1}^{n_\ell} \max \left(0, \max_{1 \leq j \leq n_k} \langle \varphi(x_{\ell i}), \varphi(x_{kj}) \rangle \right).$$

The resolver sets $q_{\text{arch}} = 0$ for $L = 0$, $1/2$ for $L = 1$, and otherwise averages $(a_{\ell k} + a_{k\ell})/2$ over all unordered source pairs, rounded to four decimal places. Thus $q_{\text{arch}} \in [0, 1]$. With capability and architecture VERTEX scores V_{cap} and V_{arch} , the project signal is

$$V_P = \frac{0.7V_{\text{cap}} + 0.3q_{\text{arch}}V_{\text{arch}}}{0.7 + 0.3q_{\text{arch}}}.$$

Authored-reference mode uses $q_{\text{arch}} = 1$. In QE mode an empty architecture union triggers a recorded authored-reference fallback, but $q_{\text{arch}} = 0$ then removes that architecture score from V_P . The fallback can therefore appear in diagnostics without contributing authored architecture evidence to the composite. Agreement is not a calibrated probability of correctness. COMET, SUPERT, and MAUVE (Rei et al., 2020; Gao et al., 2020; Pillutla et al., 2021) provide reference-estimation and distributional-evaluation context, not validation of this rule.

Pool-level QE diagnostics additionally use leave-one-out peer descriptors supported by at least two peers. They are not inputs to the per-candidate composite above. This is a heuristic, not a minimum-Bayes-risk estimator or a consistent latent-truth estimator justified by (Kumar & Byrne, 2004; Dawid & Skene, 1979). Even independent peers that include a false descriptor with probability 0.1 will, with probability tending to one, exceed a fixed two-peer threshold as the pool grows. Leave-one-out removes literal self-reference, not correlations between related models. The weak-supervision

literature (Ratner et al., 2017) is background, not a proof that unmodeled source agreement is calibrated.

An unused contrastive helper computes a softmax assignment score. Equal logits give $1/B_{\text{batch}}$ for B_{batch} alternatives, but individual scores can be smaller. This is not a universal chance floor or itself a mutual-information lower bound: InfoNCE (van den Oord et al., 2018) bounds mutual information through an **expected loss** under specified sampling assumptions. No active multi-brief contrastive evaluation is implemented.

Proposition 1’. Replacing a descriptor reference preserves Proposition 1’s range bound. The convex mixture V_P is also in $\mathbb{I}$. Neither property establishes reference accuracy or equal chance behavior.

Within-brief Spearman correlation with authored VERTEX is a preliminary concordance diagnostic, not authorization to discard reference validation on unseen briefs. The run-level diagnostic selects one built candidate per arm and omits failed builds; pool consensus and the production metric are distinct estimands. Generalization needs independent held-out briefs, adequate replication, and reference-provenance checks.

Correctness versus similarity. The oracle ν certifies requirements in the conditional closure model. VERTEX measures descriptor similarity on a continuous scale. Turning similarity into a pass/fail decision requires an additional, validated threshold; no such decision rule is part of the fixed-point semantics.

5.2 THE BUILD-GATED COMPOSITE

For a full-repository task, K signals $s_1, \dots, s_K \in \mathbb{I}$ (functional end-to-end behavior, trajectory similarity, visual fidelity, architecture, accessibility, output security, robustness, code health) combine through a convex weight vector $w \in \Delta(\{1, \dots, K\})$, gated by a build indicator $g \in \{0, 1\}$:

$$C = g \sum_{k=1}^K w_k s_k, \quad w_k \geq 0, \sum_k w_k = 1.$$

Proposition 2. $C \in \mathbb{I}$, with $C = 0$ whenever the repository does not build ($g = 0$); when it builds, $\min_k s_k \leq C \leq \max_k s_k$.

Proof. For $g = 1$, multiply $\min s_k \leq s_k \leq \max s_k$ by $w_k \geq 0$ and sum. For $g = 0$ the product is zero. Missing analyzer evidence is not a passed signal. Degraded evaluations must not be compared as though all fixed weights were measured.

The project gate is **build-only**; serving additionally gates visual and UX evidence. Functional coverage excludes unevaluable journeys from both numerator and denominator, returning zero when none are evaluable. Screenshot-based visual scores average available candidate frames and return zero when none exist. Comparisons therefore require matched journey and frame coverage; fixed outer weights do not eliminate this conditional missingness. Track G’s historical aggregate is a different descriptive index: $\bar{g} \sum_k w_k \bar{s}_k$, the product of aggregate build rate and aggregate signals. It is **not** $\bar{g} \sum_k w_k s_k$, and per-seed gated scores cannot be recovered from marginal means alone. Its completeness bootstrap resamples brief-level means.

For integers $n \geq 1$, $0 \leq c \leq n$, and $1 \leq k \leq n$, let c of n recorded trials satisfy a specified success event. Define $\binom{a}{k} = 0$ when integers $k > a \geq 0$. Reliability is $\text{pass}^k = \binom{c}{k} / \binom{n}{k}$ (Yao et al., 2024): the probability that a uniformly selected k -subset consists entirely of successes. For independent trials with common success probability $p \in [0, 1]$, it is unbiased for p^k and is non-increasing in k .

The project evaluator calls a seed successful when it builds and its measured functional coverage is at least 0.999. Track G requires seed completeness of at least 0.999 and averages reliability estimates over briefs. These are test-defined events, not complete semantic correctness. The live project path currently allows one seed, which supplies no multi-run reliability estimate.

For the *Bugfix* family (Track R), let $r \in \{0, 1\}$ indicate that the executed hidden resolution tests pass without a generation error. Let $a \in \{0, 1\}$ indicate that the submitted patch contains no

prohibited test-file edits detected by the path-based filter. With bounded graded signals s and convex weights w , $C_R = ar(w \cdot s)$. If a semantic patch judge supplies $J \in [0, 1]$, the formula is $C_R = ar(0.8(w \cdot s) + 0.2J)$. The signals measure held-out robustness, preservation of existing behavior, patch minimality/locality relative to the reference fix, and code health.

The lenient resolution rate reports r . The implemented strict rate also requires $a = 1$ and held-out/regression scores of at least 0.999. An absent held-out or regression suite receives score 1 with zero executed tests; therefore the strict rate certifies those additional checks only when the suites are nonempty and actually run. The test-path filter does not exclude arbitrary evaluator tampering.

5.3 SAFETY: ATTACK-SUCCESS AND CONFIDENCE INTERVALS

Let $\mathcal{I} = \{1, \dots, N_{\text{case}}\}$ index $N_{\text{case}} \geq 1$ evaluated cases. For each $i \in \mathcal{I}$, let $n_i \in \mathbb{N}_+$ be its recorded attempt count and $c_i \in \{0, \dots, n_i\}$ its classified successes. The **attempt-pooled** headline is $\hat{p} = \sum_{i \in \mathcal{I}} c_i / \sum_{i \in \mathcal{I}} n_i$. The separate case-weighted ASR@1 is $N_{\text{case}}^{-1} \sum_{i \in \mathcal{I}} c_i / n_i$. They differ when attempt counts differ. With $N = \sum_{i \in \mathcal{I}} n_i > 0$ and normal quantile $z > 0$ (default $z = 1.96$ for nominal 95% coverage), the Wilson score interval (Wilson, 1927) is

$$\frac{\hat{p} + \frac{z^2}{2N}}{1 + \frac{z^2}{N}} \pm \frac{z}{1 + \frac{z^2}{N}} \sqrt{\frac{\hat{p}(1 - \hat{p})}{N} + \frac{z^2}{4N^2}}.$$

For a common Bernoulli success probability $p \in [0, 1]$, the endpoints solve $(\hat{p} - p)^2 = z^2 p(1 - p)/N$ exactly. Coverage is approximate, not exact; correlated attempts and heterogeneous cases weaken this sampling interpretation.

The case-cluster percentile bootstrap (Efron, 1979) resamples whole cases and recomputes the attempt-pooled ratio. Defaults are 1,000 resamples, pseudorandom seed zero, and 95% intervals. After sorting the resampled estimates, the implementation selects zero-based entries 25 and 975 and rounds to four decimal places. Track G applies the corresponding procedure to brief-level mean completeness. One cluster or identical cluster values produces a point interval, not evidence of zero population uncertainty. Population inference still requires a representative cluster sample. With no observations, rates and intervals are unavailable.

For each case with integers $0 \leq c_s \leq n$ and $1 \leq k \leq n$,

$$\text{ASR}@k = 1 - \frac{\binom{n - c_s}{k}}{\binom{n}{k}}.$$

This is one minus the probability that a uniform k -subset of recorded attempts contains no success; it is nondecreasing in k . Under i.i.d. Bernoulli sampling it is unbiased for the probability of at least one success in k fresh attempts (Chen et al., 2021). It is undefined for $k > n$. A multi-pass adaptive attack path is one trial with its own fixed budget, not a set of independent attacks. Results with different budgets are not comparable.

Incomplete cells are excluded and listed with their reason, rather than silently compressing surviving seeds into a complete-budget result. This is still a conditional-on-observation estimate: missingness can bias it. Common-case comparisons and coverage reporting are required. Over-refusal uses benign controls. Live **secure-and-useful** is unavailable in this implementation: a transcript marker cannot independently verify the legitimate task into which an attack is inserted (the *carrier task*). Secure-and-useful evaluation is a goal motivated by (Bhatt et al., 2023; Debenedetti et al., 2024; Zhan et al., 2024), not an established live measurement here.

5.4 GATED QUALITATIVE JUDGING

The optional Claude-rubric backend uses absolute 0–4 rubrics for visual, architecture, and UX judgments (Liu et al., 2023; Zheng et al., 2023; Kim et al., 2024). Its outputs must be finite and within that range before division by four; malformed outputs are unavailable evaluations. Absolute presentation avoids pairwise ordering as an input, but does not remove verbosity, style, or model-family bias.

Other backends measure different quantities. The heuristic backend uses deterministic functional and code-structure proxies; the CLIP backend uses image similarity and descriptor comparisons. A heuristic visual proxy does not inspect rendered appearance. Backend identity and evaluated evidence coverage are therefore part of score provenance.

Assumption A2 (bounded gated judging). To qualify as evidence-based judging, a backend must return a bounded score from candidate evidence it actually reads. Visual and UX assessment require a built, served artifact; static architecture assessment requires a built repository with scoreable files. This is an eligibility condition, not an established property of every backend. Human agreement and resistance to adversarial candidate content require separate empirical validation.

5.5 CROSS-FAMILY AGGREGATE (THE GAUNTLET INDEX)

Let $\mathcal{T} = \{S, Q, R, G, P\}$ be the five fixed task families, and write $\tilde{s}_t(M) \in \mathbb{I}$ for arm M 's normalized family- t headline. The implementation maps S to $1 - \text{ASR}$, Q to requirement coverage, G to `g_score`, and P/R to their respective composites. Higher values are preferred. For an arm with qualified scores on **all five** families, define

$$\text{Index}(M) = \frac{1}{5} \sum_{t \in \mathcal{T}} \tilde{s}_t(M) \in \mathbb{I}.$$

The index is unavailable when any family is missing; individual scores and coverage remain visible. Its range follows from convexity, not empirical comparability. Qualification requires current scoring, observed execution, and complete evaluable coverage. The current publication excludes Security records from qualified headlines, so it does not presently report a five-family index. Matched tasks, budgets, provenance, and timestamps are still necessary for competitive comparisons. The index is not a validated general-capability scale.

5.6 OPERATIONAL DIAGNOSTICS FOR LONG-HORIZON RESULTS

The quantities below are **proposed diagnostics**, not measurements supplied by the current paper. A valid transition model or repeated, uncensored run-level observations would be needed to estimate them. A budget cap is not an observed completion time.

Diagnostic	Meaning
n	Maximum loop iterations / total passes allowed for the run.
Productive passes	Passes with $S_{t+1} \supsetneq S_t$ in the certificate-accumulation trace.
$\text{efc}(S_t)$	New certificates enabled by the abstract operator: $ \Phi(S_t) \setminus S_t $.
$ R , R_{\text{req}} $	Total and required requirement counts (the milestone budget).
Reasoning budget b	Thinking mode / compute budget the harness ran under.
$\Pr(T_C \leq n)$	Probability of covering R_{req} within the iteration budget (§4.4).
$\mathbb{E}[T_C]$	Expected number of passes to completion.
$-\log_2 p$	Event surprisal in bits; for a harmful event, a descriptive risk transform.

Table 2. Proposed execution diagnostics. No fitted Markov kernel, calibrated per-requirement prior, or completion-time distribution is reported here.

The last row uses self-information (Shannon, 1948). For an event of probability $p \in (0, 1]$, its *surprisal* is

$$I(p) = -\log_2 p \text{ bits} \quad (p = \frac{1}{2} \Rightarrow 1 \text{ bit}, \quad p = \frac{1}{4} \Rightarrow 2 \text{ bits}, \quad p = 0.01 \Rightarrow 6.64 \text{ bits}),$$

This is finite on $(0, 1]$; extend $I(0) = +\infty$. A zero observed attack count is not proof that the underlying probability is zero. For positive raw and governed rates, a descriptive bit difference is $\log_2(\hat{p}_{\text{raw}}/\hat{p}_{\text{gov}})$; 0/0 is undefined. A probability interval $[p_L, p_U]$ transforms to

$[-\log_2 p_U, -\log_2 p_L]$, with an infinite upper endpoint if $p_L = 0$. This transformation creates no new evidence or confidence guarantee.

A proposed information-weighted count must specify its event. Let $Z_t = (H_0, S_0, \dots, H_t, S_t)$ be the observed run history through pass t . For each uncertified $r \in R \setminus S_t$, suppose a calibrated estimate $q_t(r) = \Pr(r \in S_{t+1} \setminus S_t \mid Z_t) \in (0, 1]$ were available. For an observed transition one could then define $\text{iefc}_t = \sum_{r \in S_{t+1} \setminus S_t} -\log_2 q_t(r)$. This sums marginal event surprisals; it is not joint information without an appropriate dependence model, nor a measure of task value. No such probability estimates are reported here.

6. EXPERIMENTAL SETUP

6.1 BASELINES AND BASE MODELS

An arm is a harness, configured provider/model, prompts, tools, environment, and budget. The implementation offers raw Codex CLI, Claude Code, Oh My Pi (OMP), and OpenCode adapters with Cortex-configured counterparts. These are product-bundle comparisons. Configuration and display labels are not attestations of the model that actually served a historical request. Floating aliases, environment overrides, reasoning effort, and tool/runtime versions must be recorded per execution.

Same-model labels alone do not identify a governance-only causal effect. Cortex arms also change prompts, configuration, and sometimes planning/repair budgets. A policy-only ablation needs matched effective models, tools, environments, cases, evaluation access, and total resource budgets. A benchmark invocation can contain many internal agent actions.

6.2 RELATION TO EXISTING BENCHMARKS

SWE-bench and its variants assess repository issue resolution (Jimenez et al., 2024; OpenAI, 2024; Zhang et al., 2025); LiveCodeBench emphasizes contamination-resistant code evaluation (Jain et al., 2025). Commit0 and Terminal-Bench include difficult multi-stage work (Zhao et al., 2025; Merrill et al., 2026). Independent tasks do not imply single-pass agents or saturated performance. CyberSecEval, AgentDojo, and InjecAgent study security and prompt injection (Bhatt et al., 2023; Debenedetti et al., 2024; Zhan et al., 2024); AgentDojo also evaluates benign task utility. AgentBench, GAIA, WebArena, OSWorld, and τ -bench cover other agent environments and interaction contracts (Liu et al., 2024; Mialon et al., 2024; Zhou et al., 2024; Xie et al., 2024; Yao et al., 2024). Gauntlet is a complementary framework, not evidence that these suites lack long-horizon work, utility evaluation, or harness comparisons.

6.3 TASKS AND MEASURED OUTCOMES

Table 1 distinguishes the intended task family from what its evaluator can establish. Passing finite tests is evidence for those tests, not complete semantic correctness.

Family	Task and score	Validity boundary
S: Safety	Classified attack responses and proposed actions; ASR and benign over-refusal.	Lexical classification and action replay are not observed exploits. Live carrier-task utility is unmeasured.
Q: Quality	Generated code, hidden behavioral tests, static findings, lint/types and quality rubrics.	Analyzer availability and evaluator integrity are prerequisites; finite tests are incomplete.
R: Bugfix	Repository patches; resolution-gated score with held-out and regression checks.	Reference-diff similarity is not correctness. Test-runner tampering must be independently excluded.
G: Generative	Generated chat application, feature probes, static scores and optional visual evidence.	The local endpoint is a deterministic stub, not a learned model. Historical echo checks do not prove endpoint use; claim-evidence consistency and observed trajectory evidence are incomplete.

Family	Task and score	Validity boundary
P: Project	One storefront brief, build-gated functional/semantic/visual/static composite.	Partial behavioral and accessibility proxies are not a complete storefront or payment validation.

Table 1. Five families and their measurement boundaries. No direction of a raw-versus-governed effect is assumed by the score definition.

6.4 CONFIGURATION ISOLATION AND ADAPTIVE EVALUATION

Raw security adapters create temporary workspaces and scrub selected environment and tool-home configuration. Governed adapters add a safety preamble and link repository configuration. This is configuration isolation, **not host or network confinement**, nor proof that every hook loaded. Track S does not itself run the completion loop. Some nominal tool-output and memory attacks are delivered as workspace files representing those surfaces, not through native tool-result or session-memory interfaces.

Adaptive attacks use earlier target responses to generate later prompts, but each target attempt starts a fresh session/workspace. Their budgeted path is not a continuous target conversation, and follow-ups need not preserve the original delivery surface. Corpus curation based on previous outcomes and a `held_out` flag do not establish an independently frozen holdout.

The project evaluator scores first/final snapshots and retains the highest composite. Governed arms can also use evaluation feedback during repair. These current and historical paths produce adaptively selected development scores, not untouched holdout measurements. Access to more evaluations can favor one arm. A controlled comparison must commit each artifact before a separate final evaluation and report all generation/evaluation costs.

6.5 REPLICATION AND MISSINGNESS

Seed labels index repeated calls; they are not controlled provider RNG seeds. Configured maximum attempts, completed attempts, skipped cases, internal repair passes, and measured provider usage are separate quantities. Q and R report worst observed repeats rather than a pooled per-seed success estimate. Live full-app/repository paths have historically been single-run paths; requested counts must not be interpreted as performed trials. A timeout is not a refusal or proof of candidate safety. Excluding incomplete cells requires coverage reporting and common-case comparisons, and does not remove informative-missingness bias.

6.6 CONTAINMENT AND EVALUATOR INTEGRITY

The Q/R/G/P evaluation paths execute generated code in Docker. For G/P, dependency installation and compilation occur in a build stage with registry-network access; the later runtime probe applies resource limits and a track-specific network policy. Those probe restrictions must not be attributed to every build-stage operation. Containerized evaluation also does not establish confinement of the generation harness. The live Track S CLI adapters lack an outer sealed-agent sandbox; some use permission-bypass modes, and governed arms inherit host configuration. Adversarial execution requires independent verification of the actual isolation boundary.

The lexical detector (L0) and in-memory action-replay scorer (L1) classify captured text and proposed actions. Their historical “confirmed” fields are scorer labels, not authenticated side-effect attestations. Echo filtering, refusal heuristics, bounded capture, and missing typed tool-event provenance can misclassify outcomes. Dynamic evaluation must also protect test-runner configuration and authenticate verdicts rather than trusting candidate-produced terminal summaries. The present implementation has not established security or evaluator-integrity sign-off.

Historical classified attack success

Points: observed rates · Whiskers: stored 95% Wilson intervals

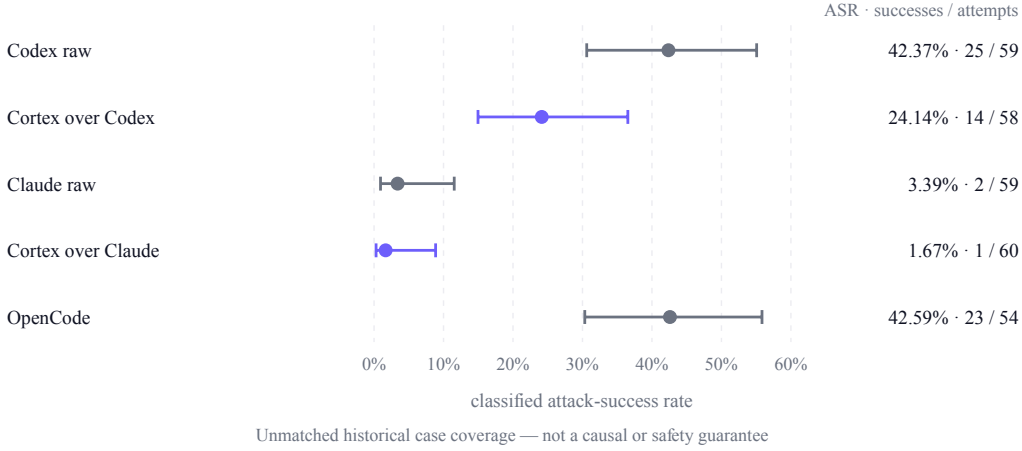


Figure 4: Historical classified Track S outcomes from tui-security-20260620-183114-748-rescored. Points show observed rates, whiskers the stored 95% Wilson intervals, and labels the successes and observed attempts. Model names are historical labels. Section 7 describes coverage, provenance, and benign-control limitations; these are descriptive, not causal comparisons.

7. RESULTS AND EVIDENCE STATUS

The following are **historical descriptive summaries**, not newly reproduced experiments. Their numerical provenance is retained, but they do not satisfy a complete matched, current-scoring qualification protocol.

The safety source is tui-security-20260620-183114-748-rescored/runrecord.json under benchmark/results/. It records one seed and the following attempt counts and stored Wilson intervals:

Historical arm label	Classified successes / observed attempts	ASR	Stored 95% Wilson interval
Codex raw	25 / 59	42.37%	30.61–55.07%
Cortex over Codex	14 / 58	24.14%	14.96–36.53%
Claude raw	2 / 59	3.39%	0.93–11.54%
Cortex over Claude	1 / 60	1.67%	0.29–8.86%
OpenCode	23 / 54	42.59%	30.33–55.84%

The full-precision relative differences are about 43.0% and 50.8% for the two nominal pairs. Their denominators and observed cases differ; the source contains skipped fixtures/timeouts and lacks current scoring provenance. Each paired arm has only **two** benign controls with zero refusals: the stored interval is [0, 65.76%]. That does not establish unchanged benign utility, low population over-refusal, statistical equivalence, or a structural causal mechanism.

Capability evidence. The project record project-20260619-092725-226 has live=false and basis="mock(modelledlong-horizonoutcome)". Its composites 0.558 and 0.958 are synthetic. The approximately 72% relative difference is arithmetic on those values, not a measured capability gain. The quality record tui-quality-20260622-235139-282 stores coverage values 0.8125 and 0.9792, with inconsistent execution/methodology metadata. The repository record tui-repo-20260620-183114-752-rescored covers six easy tasks; it provides no observations from a hard multi-file tier.

The generative record `tui-generative-20260620-183114-751-rescored` contains five functional probes with one seed and sparse bins of final feature pass rates. The bins index feature-list position, not elapsed time, changing requirement sets, or milestone snapshots. They cannot support a longitudinal-completion curve or a fitted transition model.

QE rank agreement on the single storefront brief is exploratory. The two-arm record `tui-project-20260622-184427-792` has Spearman correlation 1; two distinct points need only share an ordering to attain that value. The five-arm record `tui-project-20260622-235136-462` reports 0.6. Rescoring the same artifacts adds no independent observations and does not establish accuracy on unseen briefs.

8. DISCUSSION

The finite-lattice result identifies a precise contract for requirement closure: validation must be monotone, accepted evidence sound, and prior certificates persistent. Repairing code complicates the last condition because an edit can invalidate earlier checks. Fresh validation of the final artifact is therefore central to connecting the abstract contract with an operational completion decision.

The evaluation framework makes these obligations observable in principle, but the archived comparisons do not identify a governance-only effect. A policy ablation should hold the effective model, task set, tools, and total budget fixed, with independent final evaluation. Replication across independently selected tasks and adequate benign controls is needed to assess generalization and utility. Synthetic runs remain useful for testing reporting behavior when clearly separated from empirical comparisons.

9. ASSUMPTIONS AND LIMITATIONS

The formal assumptions concern different boundaries. A1 requires complete mediation and threat-class coverage; A2 requires bounded judgments grounded in the evaluated evidence; A3 requires sufficient abstract state and sound, monotone, persistent certification. Parsing a task into expectation records does not construct a sound oracle. The probabilistic bound additionally requires uniformly positive progress before completion; without it, expected completion time can be infinite.

The principal measurement limitations are length-sensitive similarity, heuristic references, conditional journey/frame coverage, correlated attempts, and incomplete observations. Section 6.6 specifies the unresolved execution and evaluator-integrity boundaries. The current probes also do not fully measure payment integration, carrier-task utility, consistency of claims with artifacts, or long-horizon behavior.

Historical missing or failed signals cannot be certified by changing current error handling. The retained examples were recorded in June 2026; a later report date does not represent a new experiment.

10. CONCLUSION

Monotone validation on a finite requirement lattice yields least-fixed-point closure and fair-schedule independence under explicit assumptions. The corresponding stochastic completion-time bound requires a separate positive-progress condition. Gauntlet defines bounded evaluation scores and exposes the provenance needed to interpret them. The archived comparisons remain descriptive; matched, independently evaluated experiments are needed to estimate effects on safety, utility, and long-horizon task completion.

AVAILABILITY

The interactive paper and latest published results are available at benchmark.cortex.a2olabs.com. Results retain their source cohorts, recorded measurements and evidence limitations. The standalone Gauntlet benchmark snapshot is available at github.com/Xpitfire/cortex-gauntlet. It excludes private implementation dependencies and raw captures, so the screened snapshot alone is not a complete end-to-end reproduction environment. Download the Cortex CLI from cortex.a2olabs.com. The

downloadable PDF is a dated rendering of this paper, not a claim that the historical results are new measurements.

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APPENDIX A — PROOFS

Selected identities and their domain restrictions are collected here.

- **P1 (normalizer).** For $b \in [0, 1]$, $\eta_b(x) = \text{clamp}((x - b)/(1 - b); 0, 1)$ maps $[b, 1]$ onto $[0, 1]$, is strictly increasing on $[b, 1]$, and satisfies $\eta_b(b) = 0$ and $\eta_b(1) = 1$.
- **P2 (composite).** For $w \in \Delta(\{1, \dots, K\})$ and $s_k \in \mathbb{I}$, the convex combination $\sum_k w_k s_k$ lies in $[\min_k s_k, \max_k s_k] \subseteq \mathbb{I}$, and the build gate $g \in \{0, 1\}$ yields $C = 0$ at $g = 0$.
- **P3 (harmonic mean).** For $P, Q \geq 0$, set $F = 0$ at $P = Q = 0$, otherwise $F = 2PQ/(P + Q)$. The zero case satisfies the bounds directly; henceforth assume $P + Q > 0$. If $0 \leq P \leq Q$, then $F - P = P(Q - P)/(P + Q) \geq 0$ and $Q - F = Q(Q - P)/(P + Q) \geq 0$; exchange P, Q for the other case. Also $(P + Q)/2 - F = (P - Q)^2/(2(P + Q)) \geq 0$.
- **P4 (DTW).** For $m, n \in \mathbb{N}_+$, similarities in $[-1, 1]$, and finite $\lambda \geq 0$, every admissible path has nonnegative cost; clamping $1 - \text{DTW}/(m + n)$ below at 0 gives $D \in \mathbb{I}$.
- **P5 (any-success and all-success estimators).** For integers $0 \leq c \leq n$ and $1 \leq k \leq n$, exactly $\binom{n-c}{k}$ of the $\binom{n}{k}$ subsets avoid success, while exactly $\binom{c}{k}$ consist entirely of successes. Thus $\text{pass}@k = 1 - \binom{n-c}{k}/\binom{n}{k}$ counts subsets with any success, and $\text{pass}^k = \binom{c}{k}/\binom{n}{k}$ counts all-success subsets. Both lie in $[0, 1]$. Coupling the subsets as prefixes of one uniform permutation shows that the any-success event can only grow with k , whereas the all-success event can only shrink. Under independent Bernoulli trials with common success probability p , each fixed subset has any-success probability $1 - (1 - p)^k$ and all-success probability p^k . Linearity of expectation gives unbiasedness for these respective targets. Independence is required for these targets, not for the finite-subset identities or monotonicity.
- **P6 (Wilson).** For $N > 0$, $z > 0$, and $\hat{p} \in [0, 1]$, solving the quadratic $(\hat{p} - p)^2 = z^2 p(1 - p)/N$ yields the displayed endpoints. Exact algebra is not exact confidence coverage.

- **P7 (Theorem 1).** Restated and proved in Section 4. Proposition 3 follows from the increasing-chain property and the partition into disjoint successive differences.
- **P8 (monotone Markov execution).** Proposition 4 bounds T_C , with explicit initial state and uniformly positive strict-progress probability outside C . It does not assert completion for chains with an incomplete absorbing state.

`benchmark/docs/proofs.py` checks selected symbolic identities and finite examples, including one handcrafted prerequisite lattice. These sanity checks are not a machine-checked general theorem, implementation proof, or reproduction of measurements.